Wavelet-TSLM

A new Time Series Language Model for long industrial telemetry

Fine-tuned Qwen3.5-4B that reads temporal embeddings · runs on AMD Instinct MI300X (ROCm)

Track: Intelligent Industry · Team PoliMi · Jacopo D'Abundo

What is a Time Series Language Model?

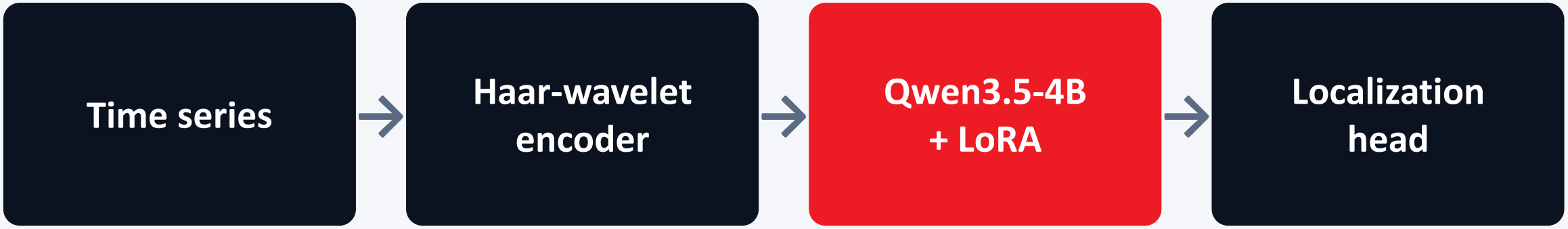
An LLM whose multimodal input is a time series, not an image or text

Until yesterday

ChatTS	OpenTSLM-Flamingo	ITFormer	Toto
Qwen3.5-4B + Chronos-2	Qwen3.5-4B + Chronos-2	Qwen3.5-4B + Chronos-2	Qwen3-VL 32B, fine-tuned

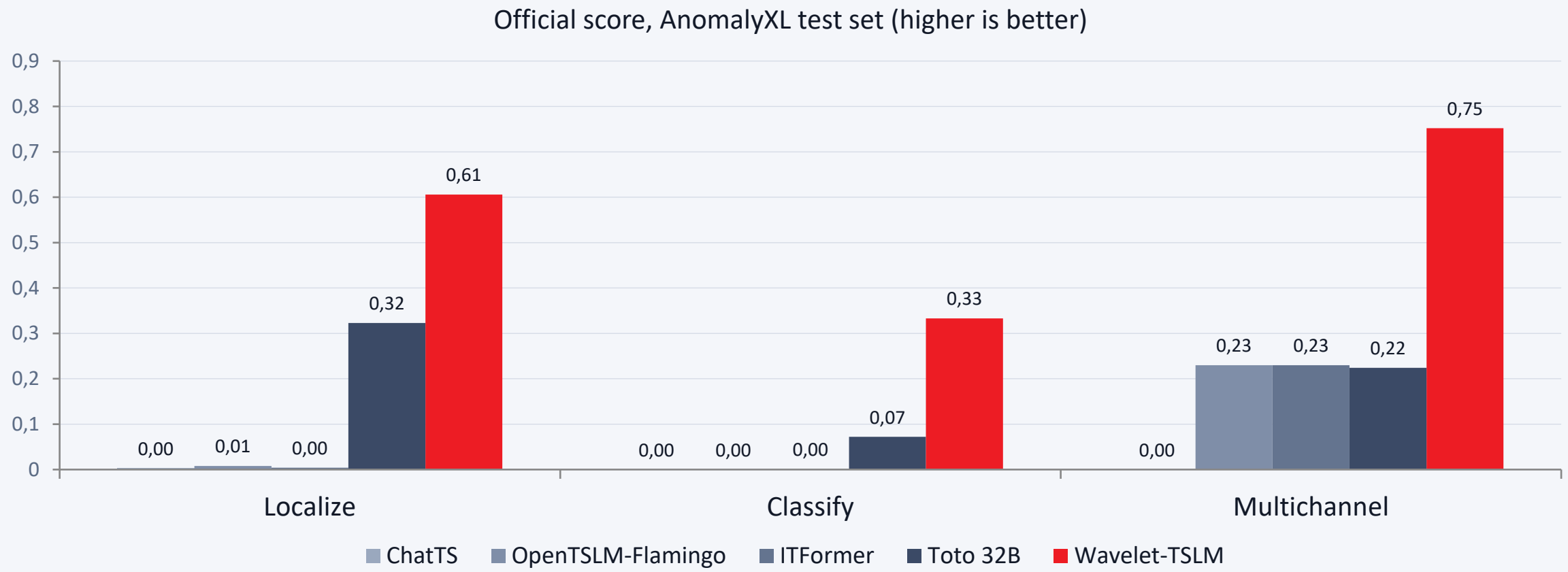
From today: Wavelet-TSLM

Wavelet-TSLM: a Qwen that reads temporal embeddings



Trained only on synthetic anomalies (AnomalyXL, ETH Zürich) · one call reads the whole window

Benchmark against the other TSLMs (ETH Zürich)



Where we do not win: magnitude (0.31 vs Toto 0.82) and lead-lag (0.25 vs 0.35–0.56). ChatTS runs out of memory on multichannel. Baselines: Table 1, TimeRLM paper (arXiv 2608.03391). Ours: one call, no zoom, same scorer.

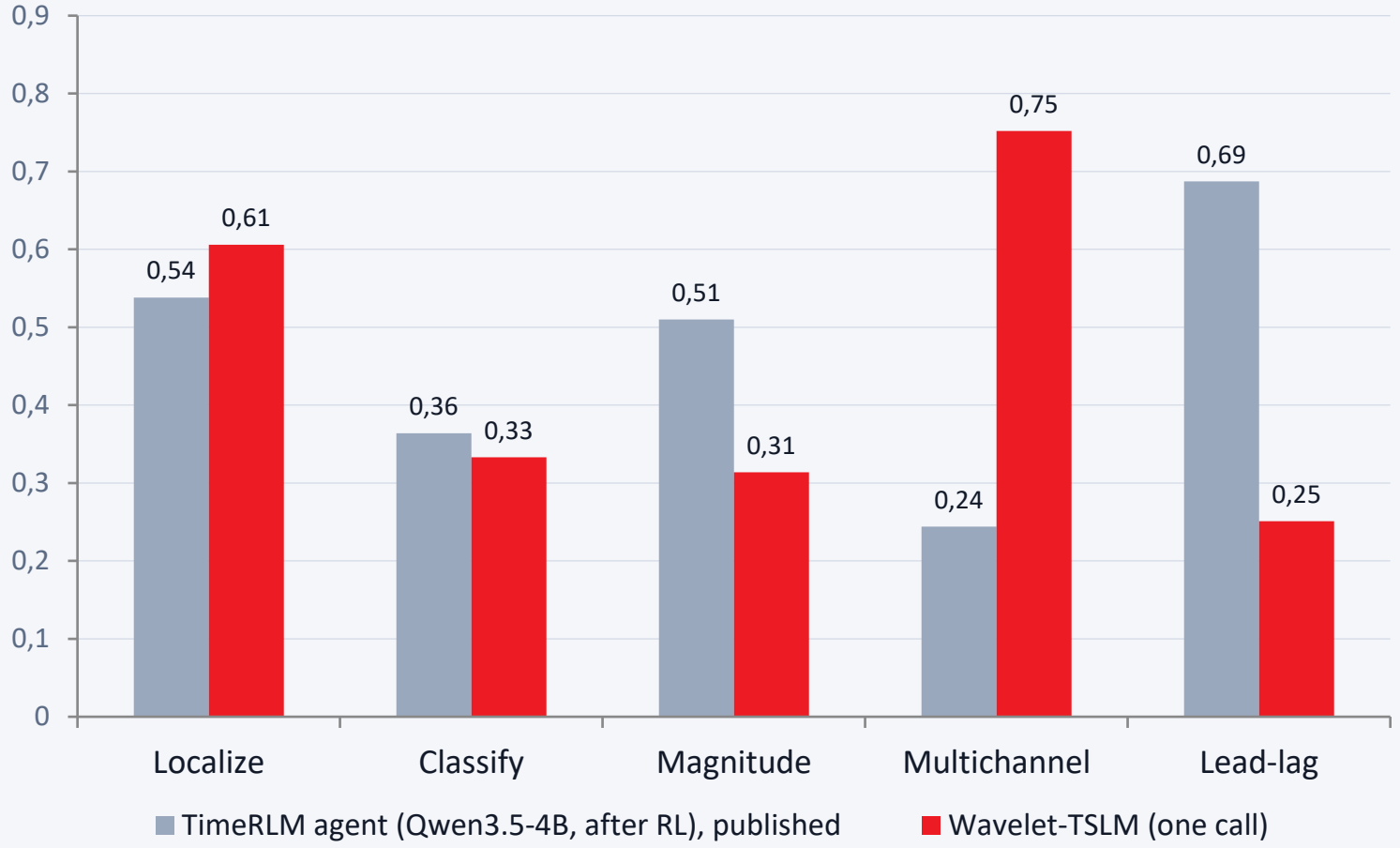
Why are the others close to zero? Input length

16×

more samples per channel
in a single call

- Wavelet-TSLM reads up to 131,072 samples per channel
- Baseline TSLMs see only the last 8,192 samples of each channel (Chronos-2 truncation)
- ChatTS runs out of memory on multichannel input
- The benchmark windows reach 131,072 samples: a design limit of those models, stated as fact

The only long-series rival: Zumarraga's agent



Not a native TSLM

A Qwen without a temporal encoder, working with Python code in a sandbox

Cost on industrial windows

Agent: about 7 turns and 5 minutes per window

Wavelet-TSLM: 1 call

Agents backed by GPT-5.5 score higher (localize 0.66–0.68)

Zero-shot on a real plant: HAI 21.03

34 / 50

attacks touched by an interval
(about 7 expected at random)

60 %

of the intervals land on an attack
(4.5 % at random)

1 call

per 9-hour, 16-channel window

Honest limit: it also flags the 18 of 18 windows with no attack, so it ranks the most anomalous stretches; it is not an alarm. On MetroPT-3 (4 slow air-leak failures) it did not transfer.

Built and measured on AMD

18.7×

less attention memory at 16k context
65.4 GiB → 3.5 GiB (forward + backward)

- PyTorch on ROCm 6.4, AMD Instinct MI300X
- FlexAttention (Triton) padded to 512
- 4 MI300X droplets on the AMD Developer Cloud, under $10 in total
- Same checkpoint, no retraining

Wavelet-TSLM: a new Time Series Language Model, on AMD. Thank you.